MARKERLESS AUTONOMOUS UAV LANDING USING VISION TRANSFORMERS FOR RISK AWARE TERRAIN SEGMENTATION

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ABSTRACT

Autonomous landing is a critical capability for unmanned aerial vehicles (UAVs), especially in GPS-denied or emergency scenarios where human intervention is limited or impossible. Traditional landing systems often rely on predefined markers or static maps, which limits their real world applicability across varying terrains and environmental conditions. To overcome these limitations, there is a growing need for intelligent systems that allow UAVs to autonomously identify safe landing zones without relying on external positioning aids.

This project presents a deep learning based approach for markerless autonomous UAV landing by leveraging Vision Transformers (ViTs) for semantic segmentation of aerial scenes. The goal is to enable drones to dynamically classify terrain into high risk and low risk areas in real time, ensuring that landing decisions are both safe and adaptable to the environment. To simulate realistic conditions, we utilize the Microsoft AirSim simulator to generate a diverse dataset that includes different terrains such as sand, grass, roads, rooftops, and water, along with variable weather conditions like fog, night, and rain, and the presence of obstacles such as rocks and power lines.

We train transformer-based segmentation models, including ViT integrated with Segmenter and Mask2Former, to interpret these complex scenes and identify safe landing zones. These models are benchmarked against conventional convolutional neural network (CNN) architectures like ResNet-based UNet. The results demonstrate that Vision Transformers not only achieve superior accuracy reaching up to 92% classification precision but also show strong generalization in previously unseen environments. This indicates their robustness in dealing with varying visual and contextual cues, which is essential for real-world deployment.

By integrating these models into a simulated UAV decision making pipeline, we enable real time terrain understanding that supports autonomous landing decisions. This approach significantly improves the reliability and flexibility of drone operations in unpredictable environments, eliminating the dependence on visual markers or GPS data. Ultimately, this project demonstrates how combining advanced vision architectures with synthetic environments can pave the way for safer and more intelligent autonomous systems.

Keywords: *Autonomous UAV Landing, Vision Transformers, Semantic Segmentation, Markerless Landing, Drone Safety, AirSim Simulation, Risk Assessment, ViT, Deep Learning, Terrain Classification.*